

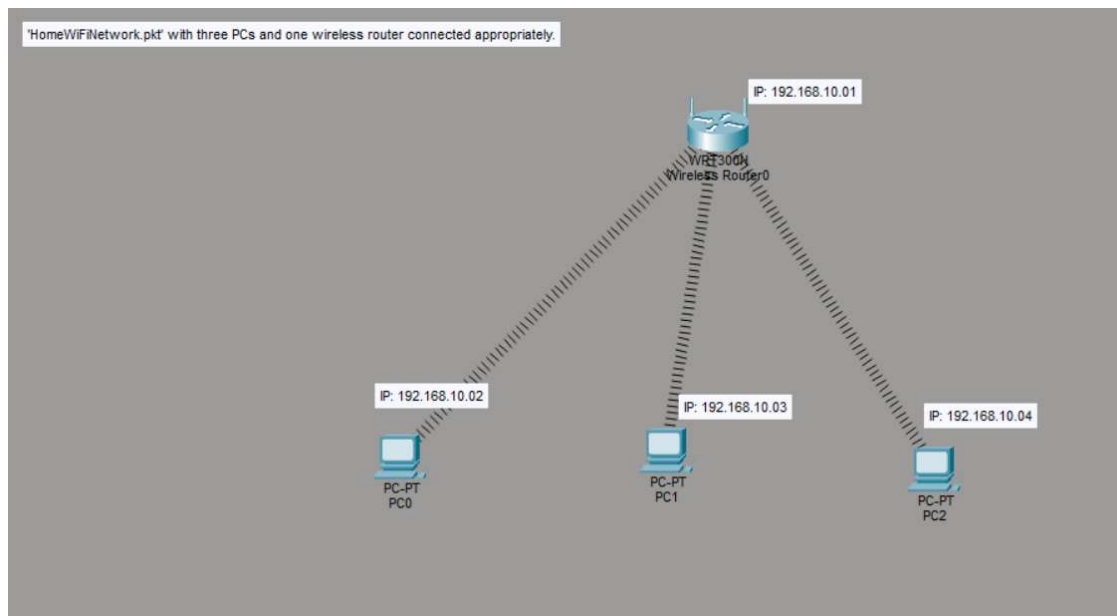
Home WiFi Network — Packet Tracer Report

Task 1. Open Cisco Packet Tracer and create a new project file named 'HomeWiFiNetwork.pkt' with three PCs and one wireless router connected appropriately.

The following devices were placed in Packet Tracer:

- 1 x Linksys WRT300N Wireless Router — labelled Wireless Router0, acting as the home WiFi access point and gateway
- 3 x PC (PC-PT) — PC0, PC1, PC2, each fitted with a PT-NM-1W wireless NIC module in place of the default wired Ethernet card

No physical cables connect the PCs to the router. Instead, each PC's wireless NIC was configured with the SSID broadcast by Wireless Router0, establishing a WiFi association shown by the dashed radio-link lines in Packet Tracer. The completed wireless topology, saved as HomeWiFiNetwork.pkt, is shown below.



screenshot 1: WRT300N Wireless Router0 and PC0–PC2 connected via WiFi

Task 2: Assign static IP addresses to each PC in your Packet Tracer network so they are on the same subnet (e.g., 192.168.10.x), and configure the default gateway to the router's IP.

All devices were placed on the 192.168.10.0/24 subnet. The WRT300N's LAN/wireless interface was set to 192.168.10.1. Each PC was configured through Desktop > PC

Wireless, first connecting to the router's SSID, then setting a static IP address, subnet mask, and default gateway under the IP Configuration section of that same window.

Device	Interface	IP Address	Subnet Mask	Default Gateway
WRT300N Wireless Router0	LAN / Wireless	192.168.10.1	255.255.255.0	—
PC0	Wireless NIC (PT-NM-1W)	192.168.10.2	255.255.255.0	192.168.10.1
PC1	Wireless NIC (PT-NM-1W)	192.168.10.3	255.255.255.0	192.168.10.1
PC2	Wireless NIC (PT-NM-1W)	192.168.10.4	255.255.255.0	192.168.10.1

Because all three PCs and the router's wireless interface share the same network ID (192.168.10.0) and subnet mask (255.255.255.0), each PC can reach the others and the router directly over WiFi, without needing a routing hop.

Task 3: Test connectivity by using the Packet Tracer 'ping' tool to verify that all PCs can communicate with each other and the router.

Connectivity was verified using the Command Prompt tool available on the Desktop tab of each PC, using the syntax: ping [other PC's IP]. Round-trip times are noticeably higher than a wired link (roughly 16–54 ms versus sub-millisecond), which is expected and confirms the traffic is genuinely passing over the simulated wireless medium rather than a wired connection.

PC0 → PC1 and PC0 → PC2

From PC0 (192.168.10.2), pinging PC1 (192.168.10.3) and PC2 (192.168.10.4) both returned 4 replies with 0% packet loss.

```
C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=51ms TTL=128
Reply from 192.168.10.3: bytes=32 time=26ms TTL=128
Reply from 192.168.10.3: bytes=32 time=36ms TTL=128
Reply from 192.168.10.3: bytes=32 time=27ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 26ms, Maximum = 51ms, Average = 35ms

C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time=42ms TTL=128
Reply from 192.168.10.4: bytes=32 time=39ms TTL=128
Reply from 192.168.10.4: bytes=32 time=36ms TTL=128
Reply from 192.168.10.4: bytes=32 time=30ms TTL=128

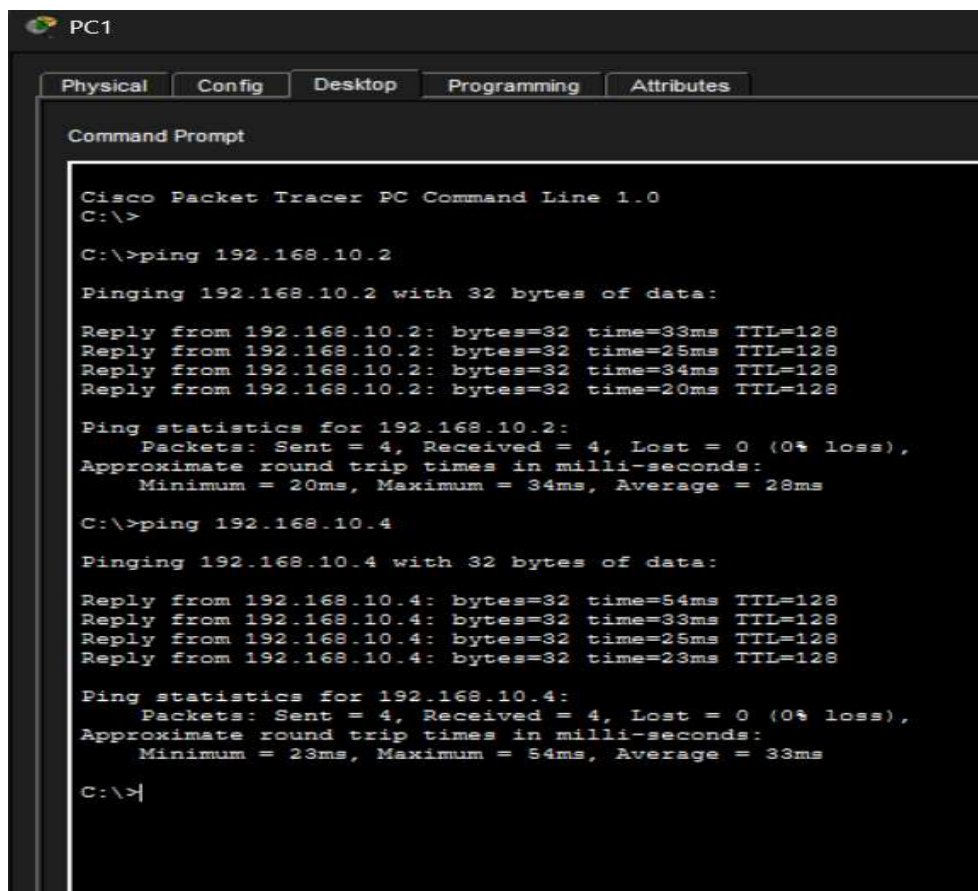
Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 30ms, Maximum = 42ms, Average = 36ms

C:\>|
```

screenshot 2: PC0 command prompt — successful wireless pings to 192.168.10.3 and 192.168.10.4

PC1 → PC0 and PC1 → PC2

From PC1 (192.168.10.3), pinging PC0 (192.168.10.2) and PC2 (192.168.10.4) both returned 4 replies with 0% packet loss.



The screenshot shows the PC1 interface in Cisco Packet Tracer. The 'Command Prompt' tab is active, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=33ms TTL=128
Reply from 192.168.10.2: bytes=32 time=25ms TTL=128
Reply from 192.168.10.2: bytes=32 time=34ms TTL=128
Reply from 192.168.10.2: bytes=32 time=20ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 34ms, Average = 28ms

C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time=54ms TTL=128
Reply from 192.168.10.4: bytes=32 time=33ms TTL=128
Reply from 192.168.10.4: bytes=32 time=25ms TTL=128
Reply from 192.168.10.4: bytes=32 time=23ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 23ms, Maximum = 54ms, Average = 33ms

C:\>|
```

screenshot 3: PC1 command prompt — successful wireless pings to 192.168.10.2 and 192.168.10.4

PC2 → PC0 and PC2 → PC1

From PC2 (192.168.10.4), pinging PC0 (192.168.10.2) and PC1 (192.168.10.3) both returned 4 replies with 0% packet loss.

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=45ms TTL=128
Reply from 192.168.10.2: bytes=32 time=25ms TTL=128
Reply from 192.168.10.2: bytes=32 time=34ms TTL=128
Reply from 192.168.10.2: bytes=32 time=16ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 16ms, Maximum = 45ms, Average = 30ms

C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=37ms TTL=128
Reply from 192.168.10.3: bytes=32 time=27ms TTL=128
Reply from 192.168.10.3: bytes=32 time=28ms TTL=128
Reply from 192.168.10.3: bytes=32 time=29ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 27ms, Maximum = 37ms, Average = 30ms

C:\>
```

screenshot 4: PC2 command prompt — successful wireless pings to 192.168.10.2 and 192.168.10.3

Every PC successfully pinged both of the other two PCs, confirming full mesh connectivity across all three hosts over WiFi.

Task 4: Document each step you took to build and configure your Packet Tracer network.

This report captures the full step by step process for HomeWiFiNetwork.pkt:

- Step 1 — Built the wireless topology: a WRT300N wireless router and three PCs, each PC upgraded with a PT-NM-1W wireless NIC and associated to the router's SSID over WiFi .
- Step 2 — Assigned static IPs 192.168.10.2–192.168.10.4 to PC0–PC2 via Desktop > PC Wireless > IP Configuration, set the WRT300N's LAN interface to 192.168.10.1, and pointed each PC's default gateway to 192.168.10.1 (see IP address table).
- Step 3 — Used the PC Command Prompt ping tool to test wireless connectivity between all three hosts; every test passed with 0% packet loss (see Figures 2–4).
- Step 4 — Compiled the wireless topology diagram, IP addressing table, and ping test results into this report.